Forensics Key

Science Olympiad North Regional Tournament at the University of Florida



Forensics Test Division C

Names:
School:
Team Number: KEY

Qualitative Analysis:

1. Of the following chemicals, which is soluble? (2 points)
   a. Calcium sulfate
   b. Cornstarch
   c. Magnesium sulfate
   d. Calcium nitrate

2. Which of the following is/are basic? (circle) (4 points)
sodium carbonate, calcium nitrate, sucrose, sodium acetate, calcium carbonate

3. Name the flame color: (6 points)
   a) NaHCO3 yellow
   b) LiCl red
   c) KCl purplish

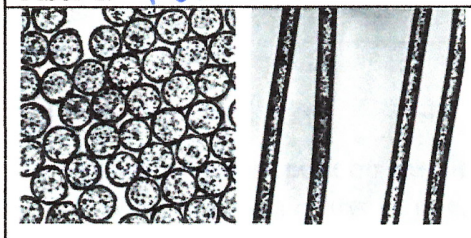
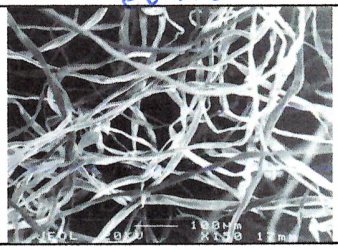
4. Why is testing for Benedict's solution useful? (3 points)
Distinguish sucrose and glucose

Polymers:
Identify the plastics from the given burn test results: (3 points per plastic)

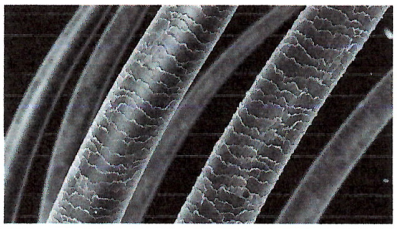
Plastic 1: yellow flame w/ green spurts, plastic doesn't drip, plastic chars, smells like hydrochloric acid, self-extinguishing PVC

Plastic 2: yellow flame, plastic drips, burns slowly, light smoke PETE (polyethylene terephthalate)

Identify the type of fiber (cotton, wool, silk, linen, nylon, spandex, or polyester) (3 points per fiber)

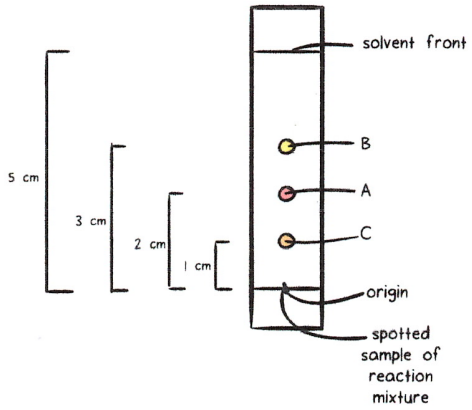
Fiber 1: POLYESTER	Fiber 2: COTTON
	

Identify the type of hair (human, bat, cow, squirrel, or horse hair) (3 points)



human hair

Chromatography:



1. Caluclate the R_f of Sample A (3 points)

$R_f = \frac{\text{distance of substance}}{\text{"solvent front}} = \frac{1\text{cm}}{5\text{cm}} = \frac{2}{5}$

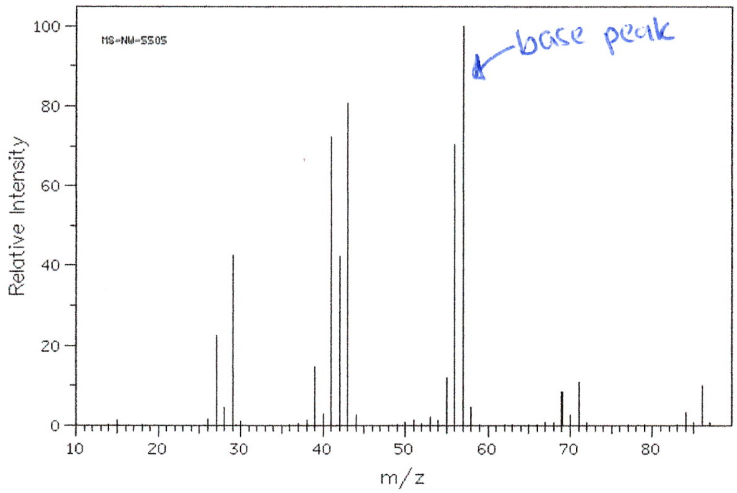
2. Caluclate the R_f of Sample B (3 points)

$R_f = \frac{3}{5}$

3. Caluclate the R_f of Sample C (3 points)

$R_f = \frac{1}{5}$

Mass Spectroscopy:



1. Label the base peak on the chart. (1 point)
2. If the ion has a charge of one, what is the molecular weight of the molecule? (3 points)

86 g/mol

3. What does TLC stand for? (2 points)

Thin layer chromatography

Glass:

Light travels from air into a shattered glass piece with an angle of incidence of 33°. The angle of refraction on the end of the glass piece is 20.997°.

1. Calculate the index of refraction (4 points)

$n_1 \sin\theta_1 = n_2 \sin\theta_2$

$n_2 = \frac{n_1 \sin\theta_1}{\sin\theta_2} = \frac{\sin 33}{\sin 20.997} = 1.5199834$

$n_1 = 1$ (air) $\quad n_2 = ?$

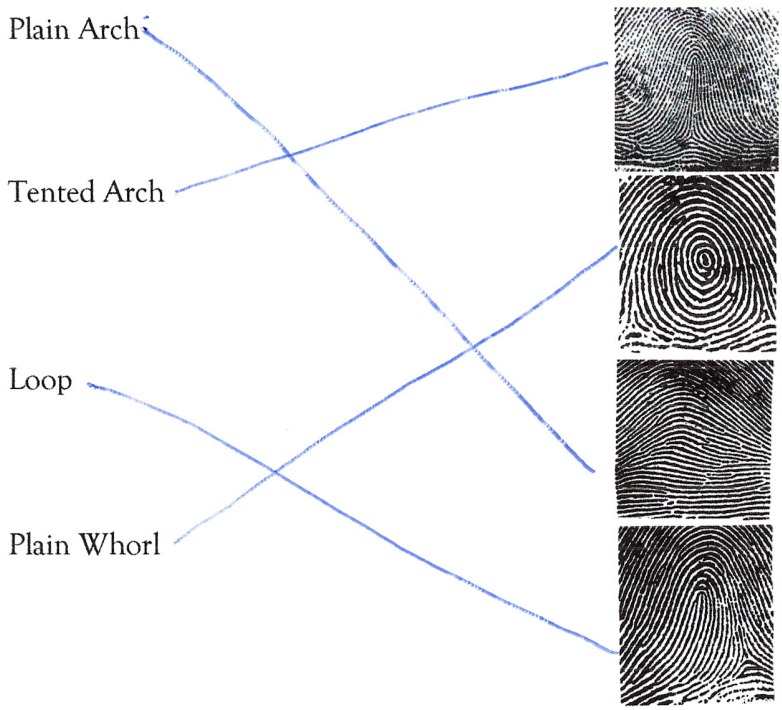
$\theta_1 = 33°$ $\quad \theta_2 = 20.997°$

2. Identify the type of glass (3 points)

Crown glass

Fingerprint:

Match the fingerprint types to the correct example (1 point each)



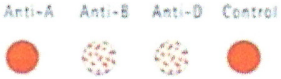
List four ways to develop fingerprints: (1 point each)

i. dusting
ii. iodine fuming
iii. ninhydrin
iv. cyanoacrylate fuming

CASE

What Happened?

Located in Ville, Bon Magasin is the best grocery store in the county. It's variety of rare products have brought people from all over the world to the tiny town. The week after Siddarth Malhotra (a Bollywood Celebrity) came to the store, Bon Magasin was robbed! When the cops arrived on the crime scene they found the following evidence:

Fingerprint	Plastic	Blood	Hair
	Burn Test Results: • Blue, yellow tipped flame • Burns slowly • Plastic drips • Has sweet odor • Floats in water PP	Test Results  B + blood	 Dog Hair

As a forensic scientist, it is your job to help the police figure out who committed this crime. The task may seem daunting, but with your skills and intelligence anything is possible!

Suspects:

1. **Vole Sain Ur**
   A college student in the town nearby
   Vole Sain Ur is a third year rubgy player (currently on probation) for Université de l'école. He has a record of being a perfect student in term of academics, but his criminal record speaks differently about his character. In the last year, he has committed an assault and has shoplifted twice.

Fingerprint	Blood Type
	O+

2. **Cam Brioleur**
   Owner of Belle Boutique, a competitor of Bon Magasin
   Cam Brioleur is the proud owner of Belle Boutique. Cam opened his store 10 years before Bon Magasin. He has worked full-time in his store ever since, not leaving any time for

family, which resulted in a divorce 2 years ago. He recently bought a dog as he was very lonely. Everyone in the town of Ville knows him as one to think of the owner of Bon Magasin as an arch nemesis. It is also well known that Cam Brioleur likes to win and has a huge crush on Siddarth Malhotra.

Fingerprint	Blood Type
	B+

3. **Charm Ante**
   The step-wife of the owner of Bon Magasin
   Charm Ante broke ties with her husband many years ago, and the divorce seemed mutual then, but recent gossip around Ville was that Charm Ante recently found out that her husband cheated on her many times during that marriage and was in the mood for revenge. Charm Ante had two twin sons the week before Siddarth Malhotra came to Ville.

Fingerprint	Blood Type
	AB-

4. **Ben Faiteur**
   A cashier at Bon Magasin
   Ben Faiteur just recently graduated from college with a degree in Finance. He worked at Bon Magasin as a cashier during his time in college, but the owner told him recently that he was going to be promoted as the Chief Financial Officer (CFO) in two weeks. Ben has an odd love for things made out of polypropylene, as one of his ex-girlfriends (a chemistry major) wrote a thesis about it.

Fingerprint	Blood Type
	O-

ANALYSIS (40 points)

Who robbed the store Bon Magasin? Why are the other suspects innocent? Be sure to support your answer with evidence and reasoning.

Theif : Cam Brioleur
- → matching finger prints + blood type.
- → dog hair

Innocent

- → Vole Sain Ur
  - → no matching evidence
  - → no motive (on probation)
- → Charm Ainte
  - → discharged w/ twins recently
  - → no matching evidence.
- → Ben Faiteur
  - → matching plastic
  - → no motive
  - → no matching evidence.

Northern Florida Regional - Repaired practice edition - Answer key

Division C | Forensics | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

Plastic-1. (4 points) Authored replacement for source item Question Plastic-1: White precipitate with NaOH dissolves in excess NaOH. Identify behavior and one ion.

answer: Amphoteric hydroxide, consistent with Al3+ as Al(OH)3 forming aluminate in excess. (4 points).

Plastic-2. (4 points) Authored replacement for source item Question Plastic-2: Chip floats in water. Which codes remain and which liquid splits PS from PET/PVC?

answer: Remain PP, LDPE, HDPE; saturated salt near 1.20 floats PS near 1.05 but PET and PVC near 1.38 still sink. (4 points).